Class 7 Notes

Quick revision: Maths, English, Science

Maths

- **Integers:** Rules: (+) × (+) = +, (-) × (-) = +, (+) × (-) = -. Division follows the same sign rules.
- **Fractions and decimals:** Multiply fractions: numerators together, denominators together. Divide by multiplying with the reciprocal. Decimal multiplication: multiply as whole numbers, then place the point.
- **Simple equations:** Solve by doing the same operation on both sides. 3x + 2 = 11, so 3x = 9, so x = 3. Transposing: change the side, change the sign.
- **Lines and angles:** Complementary angles add up to 90°, supplementary to 180°. Vertically opposite angles are equal. With parallel lines: corresponding and alternate interior angles are equal.
- **Triangles:** Angle sum = 180°. Exterior angle = sum of the two interior opposite angles. Pythagoras: hypotenuse² = base² + perpendicular² (right triangle).
- **Congruence:** Triangles are congruent if SSS, SAS, ASA or RHS holds.
- **Comparing quantities:** Profit = SP - CP, Loss = CP - SP. Profit % = (Profit / CP) × 100. Simple interest SI = (P × R × T) / 100.
- **Algebraic expressions:** Terms are added or subtracted. Like terms have the same variable and power (3x and 5x). Add like terms only.
- **Exponents:** $a^m \times a^n = a^{m+n}$. $a^m / a^n = a^{m-n}$. $(a^m)^n = a^{mn}$. $a^0 = 1$.
- **Perimeter and area:** Circle: circumference = 2πr, area = πr². Parallelogram: base × height. Triangle: 1/2 × base × height. Use π = 22/7 or 3.14.
- **Rational numbers:** Numbers of the form p/q, q not 0. Compare by cross multiplication or common denominators.
- **Data handling:** Mean = sum / number. Median = middle value of ordered data. Mode = most frequent value. Probability = favourable outcomes / total outcomes.
- **Solid shapes:** Euler's formula: F + V - E = 2 (faces + vertices - edges).

English

- **Active and passive voice:** Active: Ravi wrote the letter. Passive: The letter was written by Ravi. Passive = object + form of 'be' + past participle (+ by agent).
- **Direct and indirect speech:** Direct: She said, 'I am happy.' Indirect: She said that she was happy. Present changes to past, 'today' to 'that day', 'tomorrow' to 'the next day'.
- **Degrees of comparison:** Positive (good), comparative (better), superlative (best). Use 'than' with the comparative.
- **Clauses:** A clause has a subject and a verb. Main clause stands alone. Subordinate clause depends on a main clause (when, because, who, that).
- **Conjunctions and linkers:** although, however, therefore, moreover, unless, until, while.
- **Modals:** should (advice), must (strong duty), may (permission or possibility), could (past ability, polite request).
- **Writing:** Formal letter (to editor, principal), notice, diary entry, paragraph, message. Keep to the format and word limit.

- **Comprehension and vocabulary:** Read, underline key lines, answer in own words. Learn synonyms, antonyms, idioms and one-word substitutes.

Science

- **Photosynthesis:** Carbon dioxide + water --(sunlight, chlorophyll)--> glucose + oxygen. Leaves make food and stomata take in carbon dioxide.
- **Nutrition in animals:** Digestive system: mouth, food pipe, stomach, small intestine, large intestine. Ruminants (cow, buffalo) chew the cud.
- **Heat:** Conduction (solids), convection (liquids and gases), radiation (no medium). Normal body temperature is 37°C. Clinical thermometer range 35°C to 42°C.
- **Acids, bases, salts:** Acids turn blue litmus red; bases turn red litmus blue. Acid + base → salt + water (neutralisation). Natural indicators: turmeric, china rose, red cabbage.
- **Physical and chemical changes:** Physical changes can be reversed (melting, dissolving). Chemical changes form new substances (burning, rusting, cooking).
- **Respiration:** Glucose + oxygen → carbon dioxide + water + energy. Anaerobic respiration happens without oxygen (yeast, muscles during heavy exercise).
- **Transportation:** Blood: red cells (carry oxygen), white cells (fight germs), platelets (clotting), plasma. Xylem carries water in plants, phloem carries food. Transpiration is loss of water vapour from leaves.
- **Reproduction in plants:** Asexual: vegetative propagation, budding, fragmentation, spores. Sexual: flowers (pollination, fertilisation, seed formation).
- **Motion and time:** Speed = distance / time. Unit: m/s or km/h. Uniform motion: equal distances in equal times.
- **Electric current:** Effects of current: heating, magnetic (electromagnet). An electric fuse melts and breaks the circuit when current is too high.
- **Light:** Law of reflection: angle of incidence = angle of reflection. A plane mirror makes an erect image of the same size, laterally inverted.
- **Soil and water:** Soil layers: A (topsoil), B, C, bedrock. Loamy soil is best for crops. Save water and recharge groundwater.
- **Weather and winds:** Warm air rises, cool air moves in and makes wind. Cyclones form over warm oceans. Climate is the long-term pattern of weather.